

Connecting the two worlds

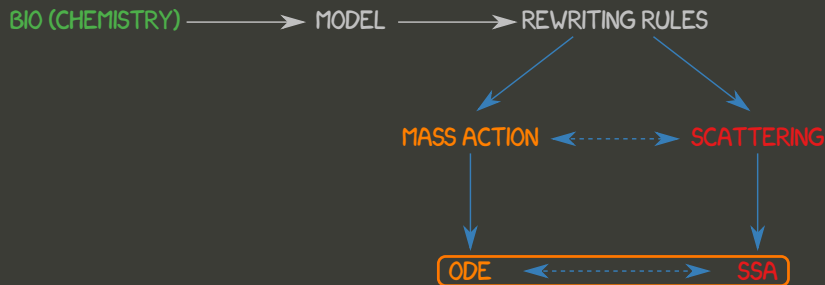
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Connecting the two worlds



ODE Vs SSA

$$\frac{dx_j}{dt} = \sum_{\mu}^M k_{\mu} \delta_{\mu,j} \prod_p^N [x_p]^{\alpha_{\mu,p}}$$

$$P(\mu, \tau) \propto a_{\mu} = c_{\mu} h_{\mu}$$

ODE Vs SSA

$$\frac{d[x_j]}{dt} = \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \prod_p^N [x_p]^{\alpha_{\mu,p}}$$

$$P(\mu, \tau) \propto a_{\mu} = c_{\mu} h_{\mu}$$

Average reaction rate per unit time

ODE Vs SSA

$$[x_p] = \frac{\langle x_p \rangle}{V} = \frac{X_p}{N_A V} \quad [mol\ l^{-1}]$$

ODE Vs SSA

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??? c_μ Vs. k_μ

ODE Vs. SSA

$$\frac{d[x_j]}{dt} = \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \prod_p^N [x_p]^{\alpha_{\mu,p}}$$

$$[x_p] = \frac{X_p}{N_A V}$$

ODE Vs. SSA

$$\frac{d[x_j]}{dt} = \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \prod_p^N [x_p]^{\alpha_{\mu,p}} \quad [x_p] = \frac{X_p}{N_A V}$$

averaging over many samples $\langle \rangle$

ODE Vs. SSA

$$\frac{d\langle [x_j] \rangle}{dt} =$$

ODE Vs. SSA

$$\begin{aligned}\frac{d\langle [x_j] \rangle}{dt} &= \\ &= \frac{d}{dt} \frac{X_j}{N_A V} =\end{aligned}$$

ODE Vs. SSA

$$\begin{aligned}\frac{d\langle [x_j] \rangle}{dt} &= \\ &= \frac{d}{dt} \frac{X_j}{N_A V} = \frac{1}{N_A V} \frac{dX_j}{dt} =\end{aligned}$$

ODE Vs. SSA

$$\begin{aligned}\frac{d\langle [x_j] \rangle}{dt} &= \\&= \frac{d}{dt} \frac{X_j}{N_A V} = \frac{1}{N_A V} \frac{dX_j}{dt} = \\&= \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \prod_p^N [x_p]^{\alpha_{\mu,p}}\end{aligned}$$

ODE Vs. SSA

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ODE Vs. SSA

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ODE Vs. SSA

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ODE Vs. SSA

$$\begin{aligned}\frac{d\langle [x_j] \rangle}{dt} &= \\&= \frac{d}{dt} \frac{X_j}{N_A V} = \frac{1}{N_A V} \frac{dX_j}{dt} = \\&= \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \prod_p^N [x_p]^{\alpha_{\mu,p}} = \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \prod_p^N \left(\frac{X_p}{N_A V} \right)^{\alpha_{\mu,p}} \\&= \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \prod_p^N \frac{X_p^{\alpha_{\mu,p}}}{(N_A V)^{\alpha_{\mu,p}}} = \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \frac{\prod_p^N X_p^{\alpha_{\mu,p}}}{\prod_p^N (N_A V)^{\alpha_{\mu,p}}} \\&= \sum_{\mu}^M \delta_{\mu,j} k_{\mu} \frac{\prod_p^N X_p^{\alpha_{\mu,p}}}{(N_A V)^{\sum_p^N \alpha_{\mu,p}}}\end{aligned}$$

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ODE Vs. SSA

$$\frac{1}{N_A V} \frac{dX_j}{dt} = \sum_{\mu}^M \delta_{\mu,j} \frac{k_{\mu}}{(N_A V) \sum_p^N \alpha_{\mu,p}} \prod_p^N x_p^{\alpha_{\mu,p}}$$

ODE Vs. SSA

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$$\frac{dX_j}{dt} = \sum_{\mu}^M \delta_{\mu,j} \frac{k_{\mu}}{(N_A V) (\sum_p^N \alpha_{\mu,p})^{-1}} \prod_p^N x_p^{\alpha_{\mu,p}}$$

ODE Vs. SSA

$$\frac{1}{N_A V} \frac{dX_j}{dt} = \sum_{\mu}^M \delta_{\mu,j} \frac{k_{\mu}}{(N_A V)^{\sum_p^N \alpha_{\mu,p}}} \prod_p^N x_p^{\alpha_{\mu,p}}$$

$$\frac{dX_j}{dt} = \sum_{\mu}^M \delta_{\mu,j} \frac{k_{\mu}}{(N_A V)^{(\sum_p^N \alpha_{\mu,p})-1}} \prod_p^N x_p^{\alpha_{\mu,p}}$$

$$\frac{dX_j}{dt} = \sum_{\mu}^M \delta_{\mu,j} \frac{k_{\mu}}{(N_A V)^{K_{\mu}-1}} \prod_p^N x_p^{\alpha_{\mu,p}} \quad K_{\mu} = \sum_p^N \alpha_{\mu,p}$$

ODE Vs. SSA

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ODE Vs. SSA

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ODE Vs. SSA

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$$k'_{\mu} = \frac{k_{\mu}}{(N_A V)^{K_{\mu}-1}}$$

ODE Vs. SSA: average reaction rate per unit time

$$\frac{dX_j}{dt} = \sum_{\mu} \delta_{\mu,j} k'_{\mu} \prod_p x_p^{\alpha_{\mu,p}}$$

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$$h_\mu = \prod_p^N \binom{X_p}{\alpha_{\mu,p}}$$

ODE Vs. SSA

$$P(\mu, \tau) \propto \mathbf{a}_\mu = \mathbf{c}_\mu \mathbf{h}_\mu$$

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ODE Vs. SSA

$$P(\mu, \tau) \propto \mathbf{a}_\mu = \mathbf{c}_\mu \mathbf{h}_\mu$$

$$h_\mu = \prod_p^N \binom{X_p}{\alpha_{\mu,p}} \propto \prod_p^N \frac{X_p^{\alpha_{\mu,p}}}{\alpha_{\mu,p}!}$$

$$\mathbf{a}_\mu \propto \mathbf{c}_\mu \frac{\prod_p^N X_p^{\alpha_{\mu,p}}}{\prod_p^N \alpha_{\mu,p}!}$$

ODE Vs. SSA

The average rate $\langle r_\mu \rangle$ at which reaction r_μ occurs in V per unit time over many samplings is

$$\langle r_\mu \rangle = \langle a_\mu \rangle = c_\mu \frac{\left\langle \prod_p^N x_p^{\alpha_{\mu,p}} \right\rangle}{\prod_p^N \alpha_{\mu,p}!}$$

ODE Vs. SSA

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The average reaction rate at which reaction r_μ occurs in V per unit time is

$$\langle r_\mu \rangle = k'_\mu \left\langle \prod_p^N x_p^{\alpha_{\mu,p}} \right\rangle$$

ODE Vs. SSA

$$\frac{c_\mu}{\prod_p^N \alpha_{\mu,p}!} \left\langle \prod_p^N x_p^{\alpha_{\mu,p}} \right\rangle = k'_\mu \left\langle \prod_p^N x_p^{\alpha_{\mu,p}} \right\rangle$$

ODE Vs. SSA

$$\frac{c_\mu}{\prod_p^N \alpha_{\mu,p}!} \left\langle \prod_p^N x_p^{\alpha_{\mu,p}} \right\rangle = k'_\mu \left\langle \prod_p^N x_p^{\alpha_{\mu,p}} \right\rangle$$

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ODE Vs. SSA

$$\frac{c_\mu}{\prod_p^N \alpha_{\mu,p}!} \left\langle \prod_p^N x_p^{\alpha_{\mu,p}} \right\rangle = k'_\mu \left\langle \prod_p^N x_p^{\alpha_{\mu,p}} \right\rangle$$

$$\frac{c_\mu}{\prod_p^N \alpha_{\mu,p}!} = k'_\mu$$

$$\frac{c_\mu}{\prod_p^N \alpha_{\mu,p}!} = \frac{k_\mu}{(N_A V)^{K_\mu - 1}}$$

ODE Vs. SSA

$$k_{\mu} = c_{\mu} \frac{(N_A V)^{K_{\mu}-1}}{\prod_p^N \alpha_{\mu,p}!}$$

ODE Vs. SSA

$$k_{\mu} = c_{\mu} \frac{(N_A V)^{K_{\mu}-1}}{\prod_p^N \alpha_{\mu,p}!}$$

$$K_{\mu} = \sum_p^N \alpha_{\mu,p}$$

$$[x_p] = \frac{X_p}{N_A V}$$

the Brusselator Dynamics: ODE & Stochastic

